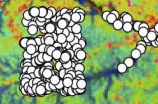
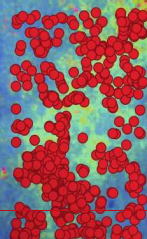


Sampling and overfitting

AI for ecologists

Paul Tresson



Introduction

What do we want when modelling ?

- Understand things

What do we want when modelling ?

- Understand things
- Predict things

What do we want when modelling ?

“All models are wrong, but some are useful”

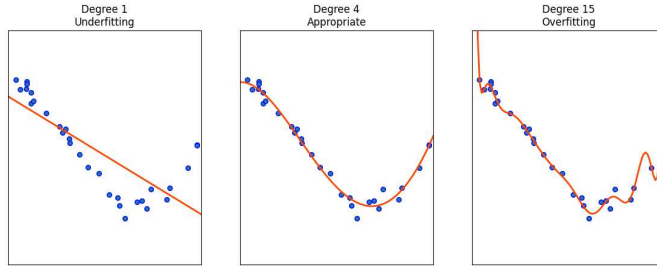
George E. P. Box

What do we want when modelling ?

- **Robustness:** Useful when mistakes
- **Generalization:** Useful applied elsewhere

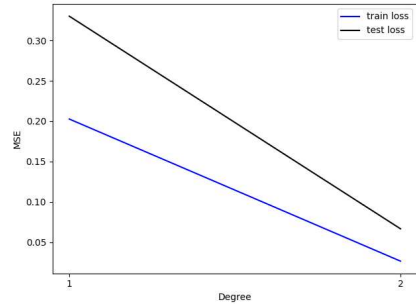
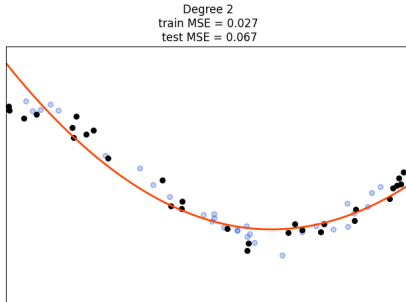
Overfitting

What is overfitting

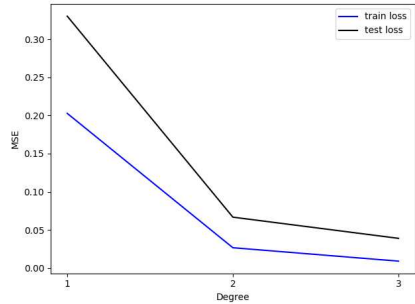
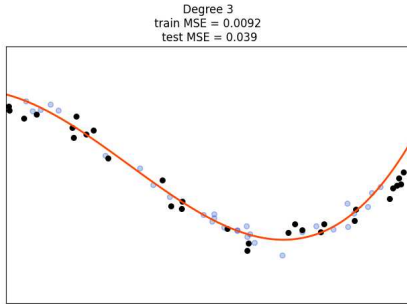


adapted from scikit-learn docs

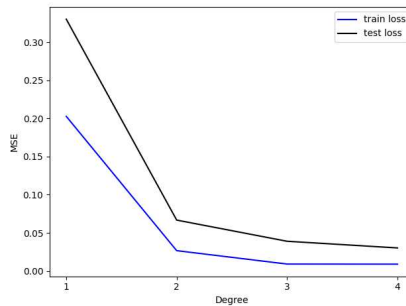
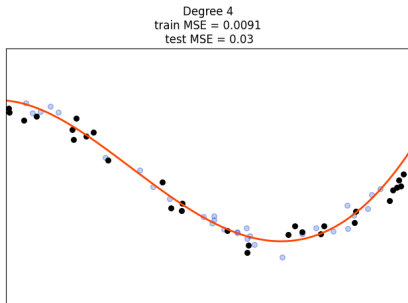
Common tools and intuitions - Train/Test loss



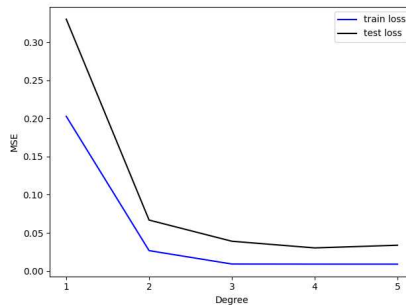
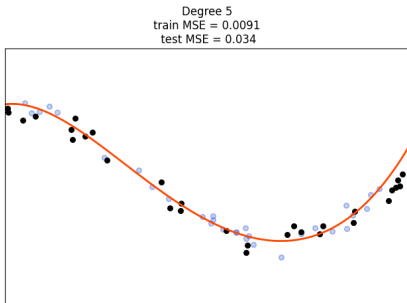
Common tools and intuitions - Train/Test loss



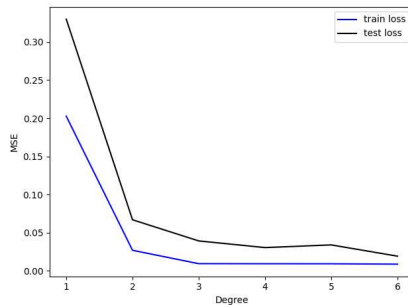
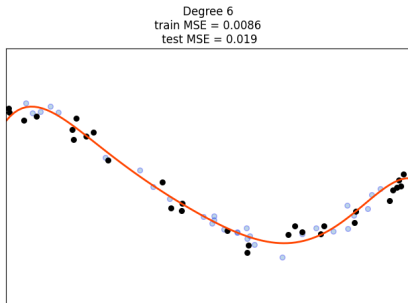
Common tools and intuitions - Train/Test loss



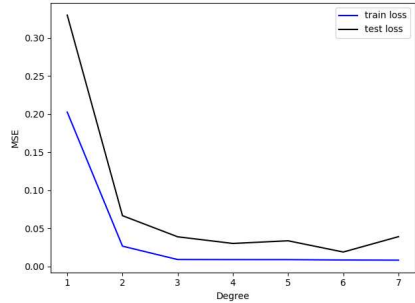
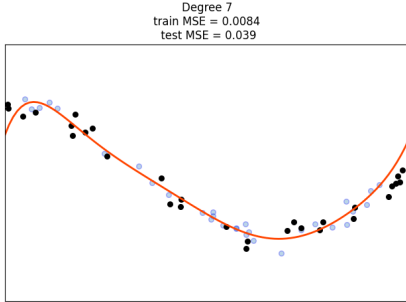
Common tools and intuitions - Train/Test loss



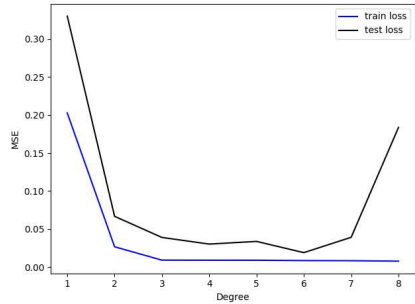
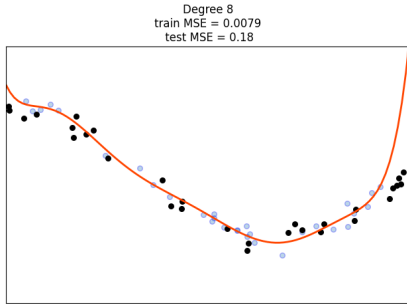
Common tools and intuitions - Train/Test loss



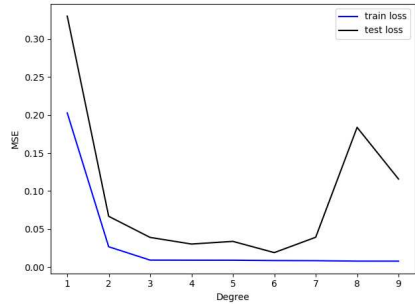
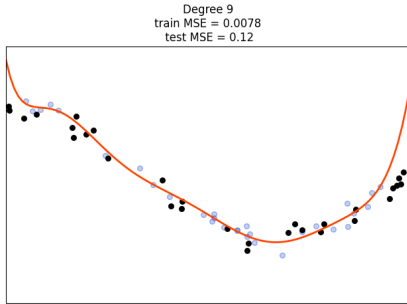
Common tools and intuitions - Train/Test loss



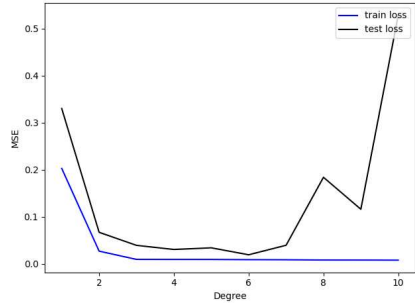
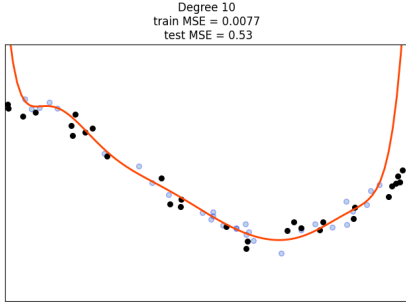
Common tools and intuitions - Train/Test loss



Common tools and intuitions - Train/Test loss



Common tools and intuitions - Train/Test loss



Common tools and intuitions - Train/Test loss

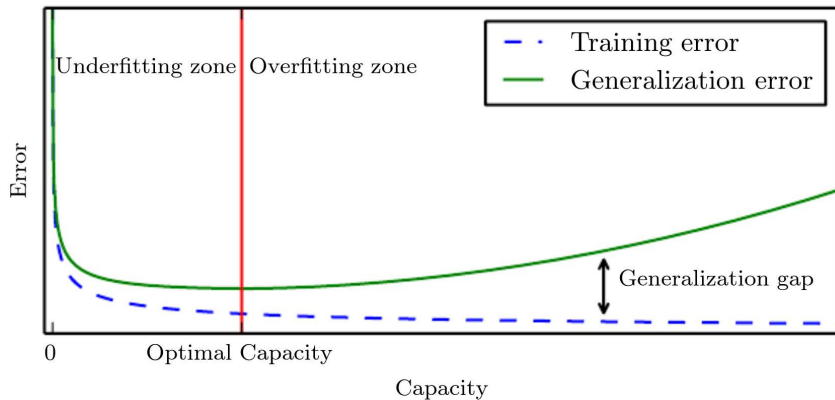


Figure from Goodfellow et al., 2016

Common tools and intuitions - AIC/BIC

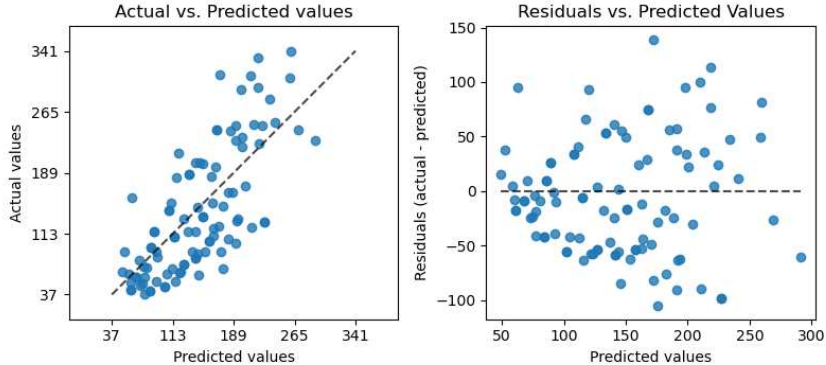
Akaike information criterion (AIC)

Bayesian information criterion (BIC)

Is the model parameter efficient ?

Common tools and intuitions - Biases

Plotting cross-validated predictions



from scikit-learn docs

And in Machine(/Deep) Learning ??

How many parameters to have
Shrek learning botany starting from random noise ?

And in Machine(/Deep) Learning ??



$\approx 2.5B$?

Root Causes

Too many parameters

Root Causes

Too many parameters

Too little training data

Root Causes

Too many parameters

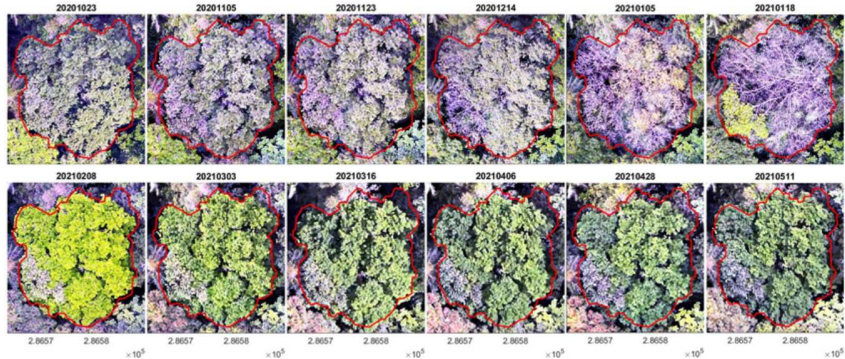
Too little training data

(bad) training data

Illustrated examples in Ecology

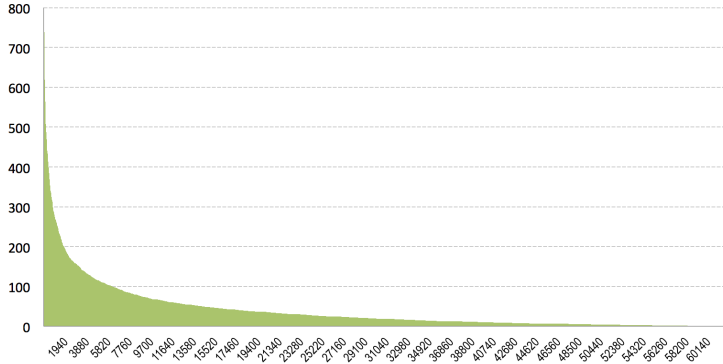
Constraints in ecology

Data from the real world is noisy,



Constraints in ecology

Data from the real world is noisy, unbalanced,



Constraints in ecology

Data from the real world is noisy, unbalanced, hard to collect,



Constraints in ecology

Data from the real world is noisy, unbalanced, hard to collect, hard to interpret.



Constraints in ecology

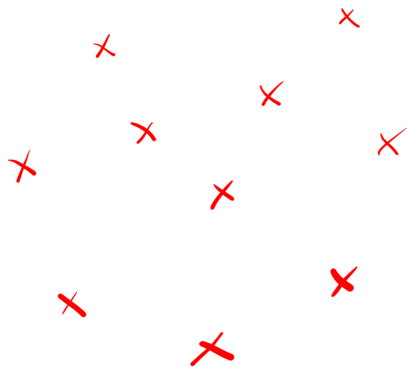
Data from the real world is noisy, unbalanced, hard to collect, hard to interpret.



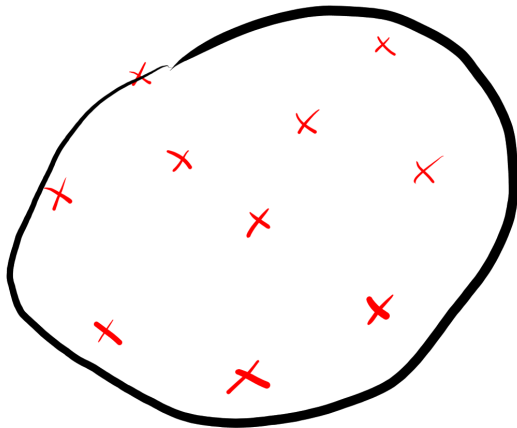
Constraints in ecology

Data from the real world is noisy, unbalanced, hard to collect, hard to interpret.

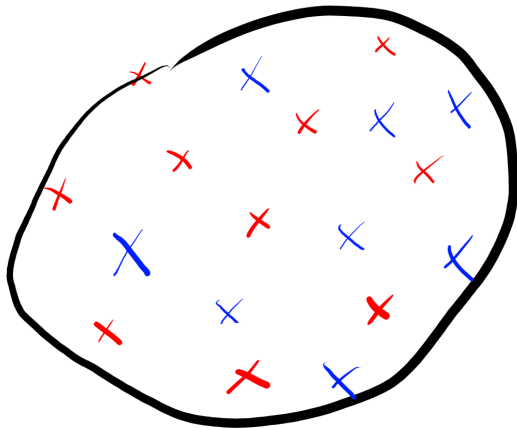




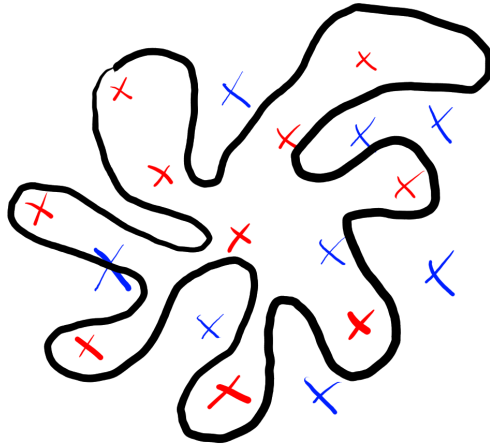
Train set



A good fitted model



Test set

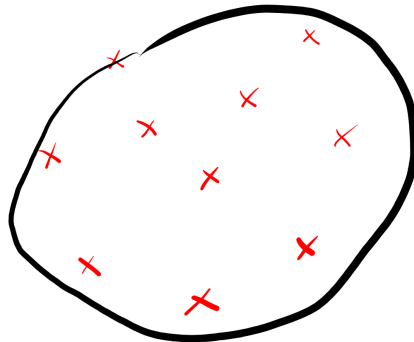


An overfitted model

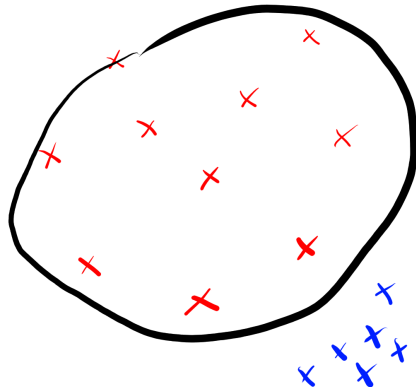
Biases in the train set



Biases in the train set



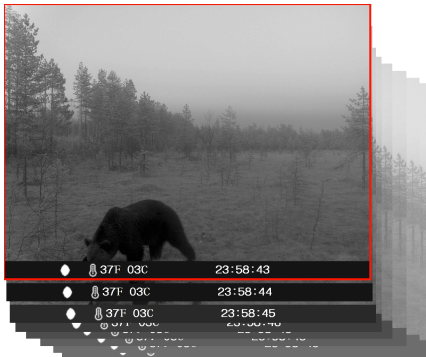
Biases in the train set



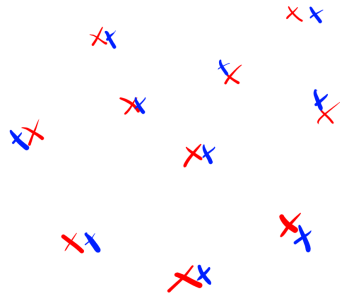
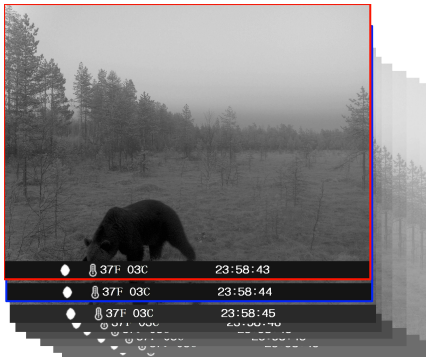
Biases in the train set - autocorrelation



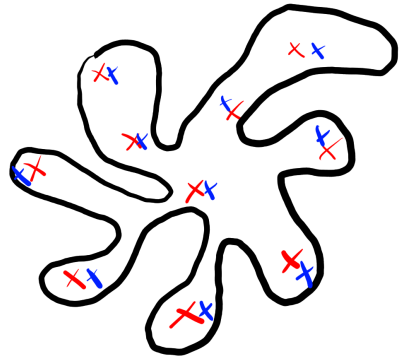
Biases in the train set - autocorrelation



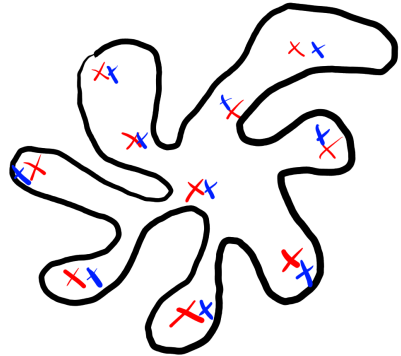
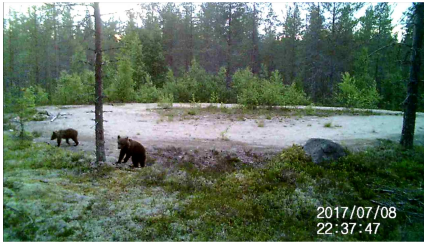
Biases in the train set - autocorrelation



Biases in the train set - autocorrelation



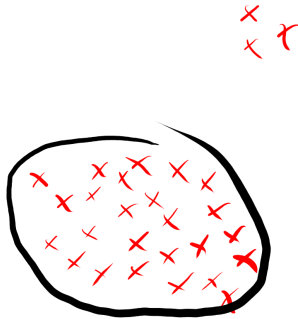
Biases in the train set - autocorrelation



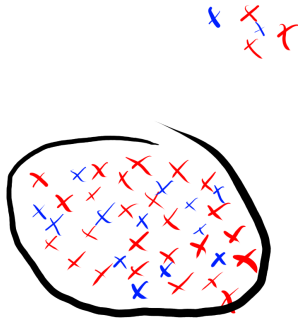
Unbalanced data



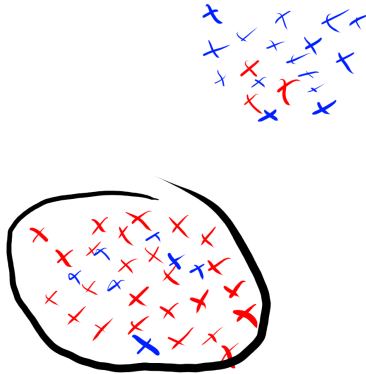
Unbalanced data



Unbalanced data



Unbalanced data



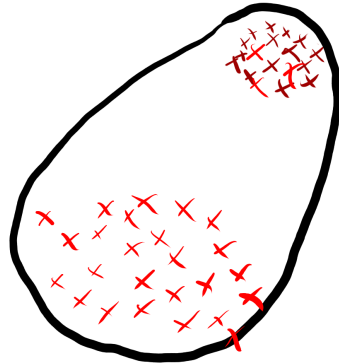
Deal with unbalanced data

- Oversample ?



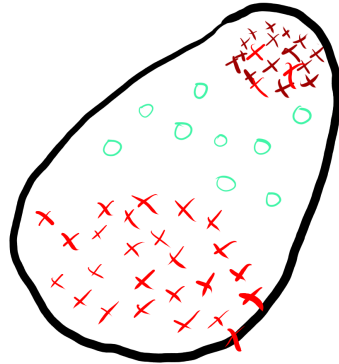
Deal with unbalanced data

- Oversample ?



Deal with unbalanced data

- Oversample ?



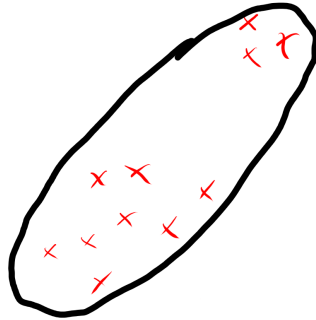
Deal with unbalanced data

- Oversample ?
- Undersample/saturate ?



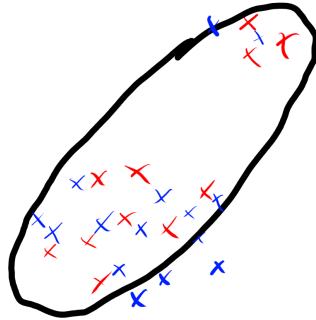
Deal with unbalanced data

- Oversample ?
- Undersample/saturate ?



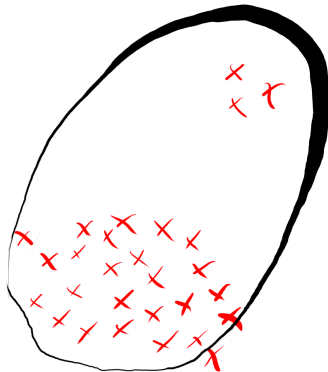
Deal with unbalanced data

- Oversample ?
- Undersample/saturate ?



Deal with unbalanced data

- Oversample ?
- Undersample/saturate ?
- Adapt loss ?



Deal with lack of data

- Data augmentation



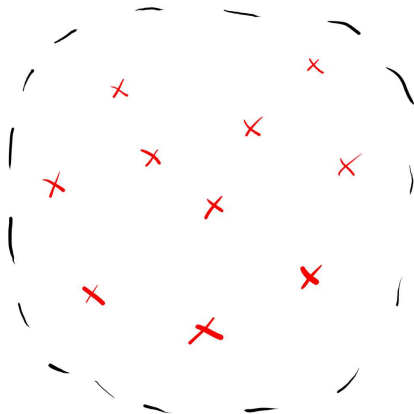
Deal with lack of data

- Data augmentation



Deal with lack of data

- Data augmentation
- Pretrained model



Deal with lack of data

- Data augmentation
- Pretrained model
- ... **collect more data**

Play with your model

- Dropout
- Pruning
- Ablation studies
- Distillation
- Ensembles

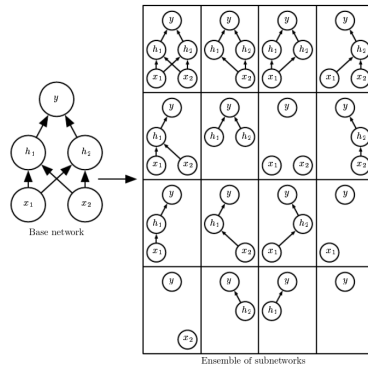


Figure from Goodfellow et al., 2016

Need to be very careful on how to evaluate

How to sample and evaluate ?

Random split ?

“random split training validation 80/20”

Random split ?

“random split training validation 80/20”

For the uncured dataset, we randomly sample 142 million images

Oquab et al., 2023

Random split ?

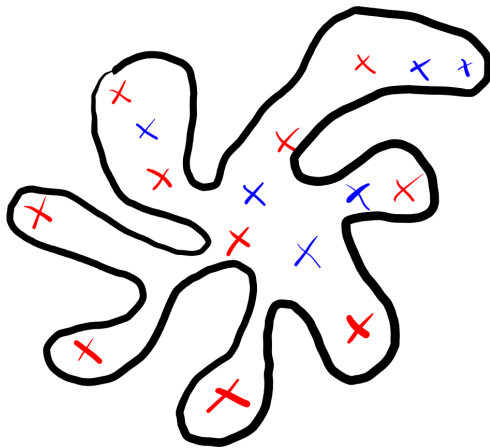
“random split training validation 80/20”

For the uncured dataset, we randomly sample 142 million images

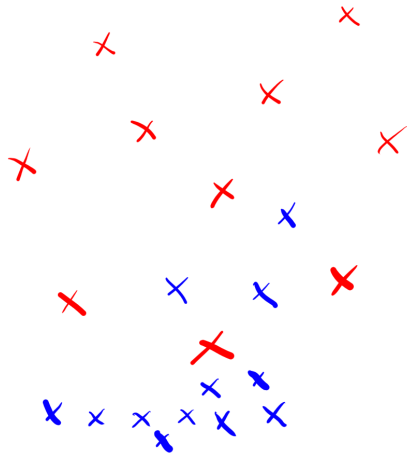
Oquab et al., 2023

Works for huge DL papers, maybe not for you

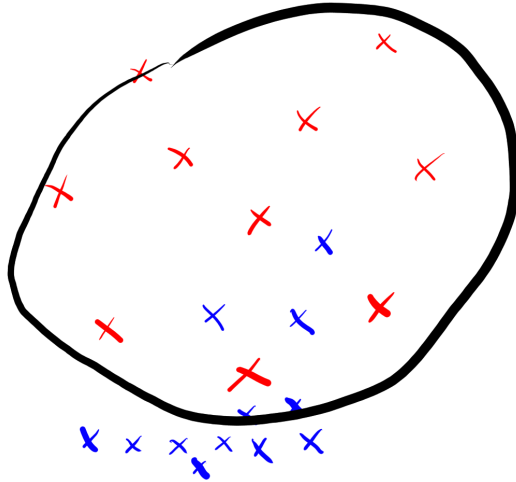
Overfitting the test set



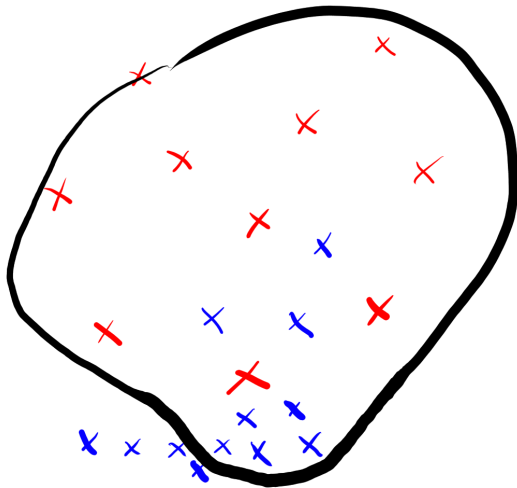
Overfitting the test set



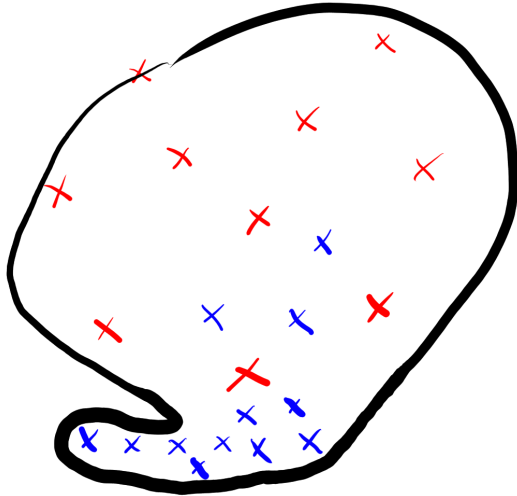
Overfitting the test set



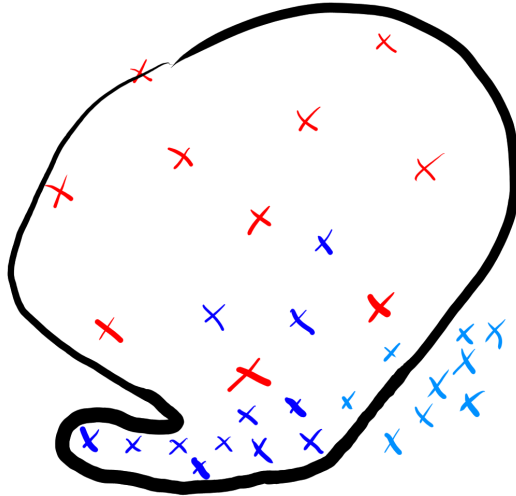
Overfitting the test set



Overfitting the test set



Overfitting the test set



Cross-validation

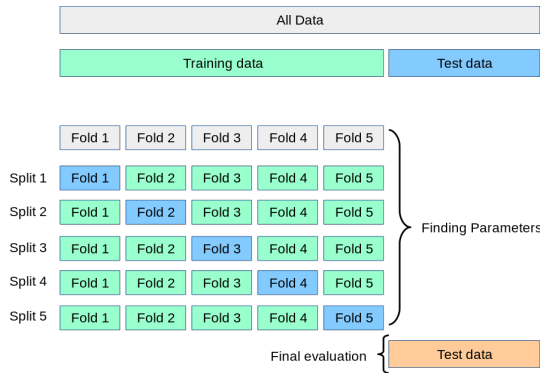


Figure from scikit-learn docs

Cross-validation

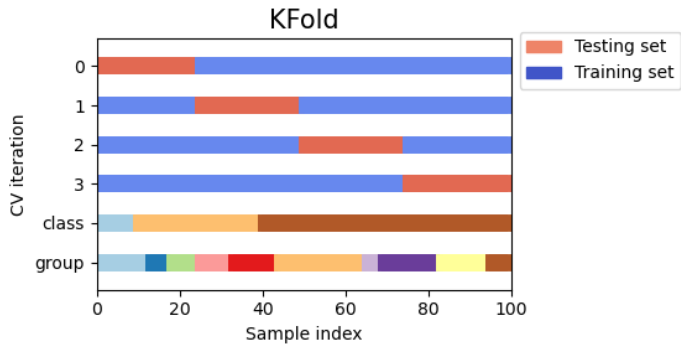


Figure from scikit-learn docs

Cross-validation

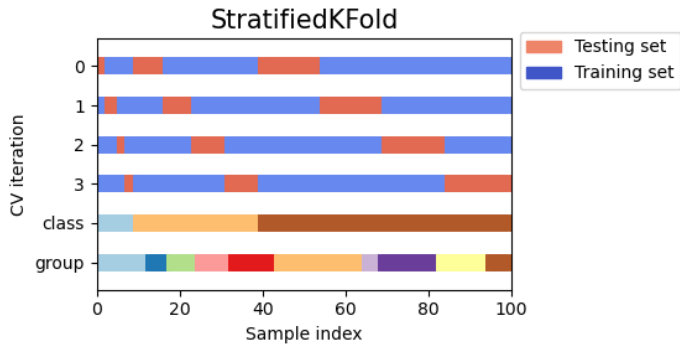


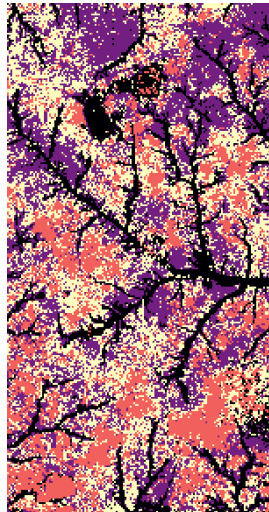
Figure from scikit-learn docs

Case studies

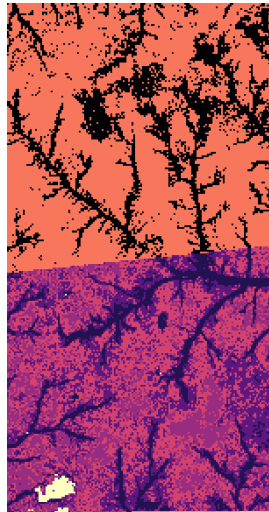
Case study : Spatial cross-validation



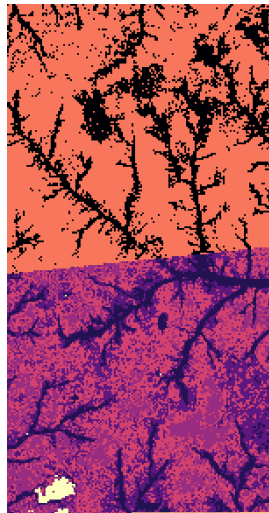
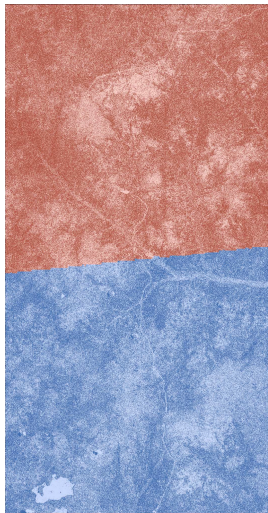
Case study : Spatial cross-validation



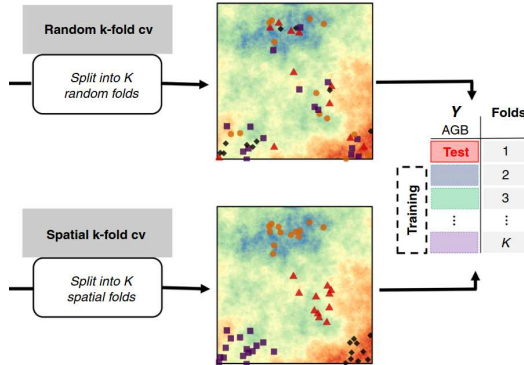
Case study : Spatial cross-validation



Case study : Spatial cross-validation

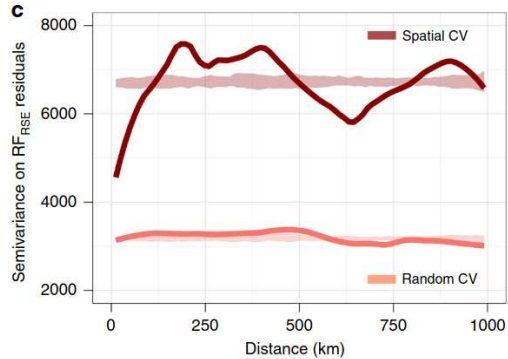


Case study : Spatial cross-validation



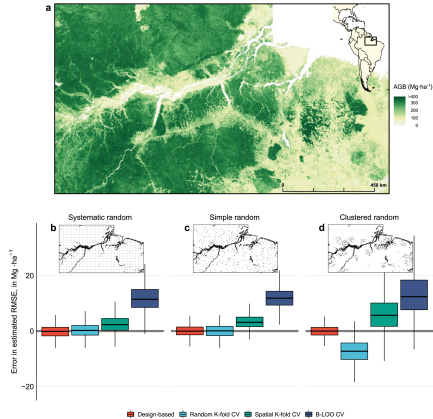
See. Ploton et al., 2020

Case study : Spatial cross-validation



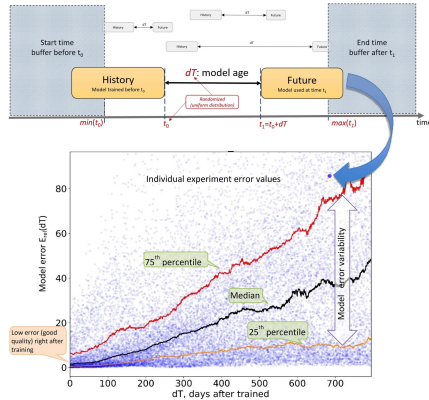
See. Ploton et al., 2020

Case study : Spatial cross-validation



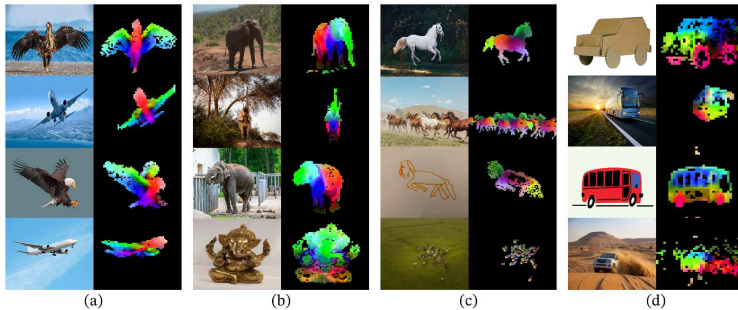
See. Wadoux et al., 2021

Case study : Aging models ?



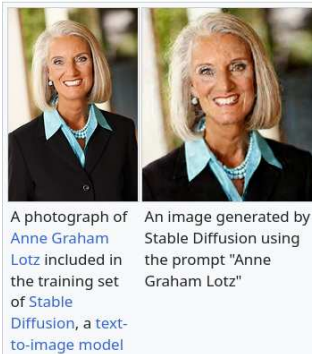
See. Vela et al., 2022

Perspective : Foundation models ?

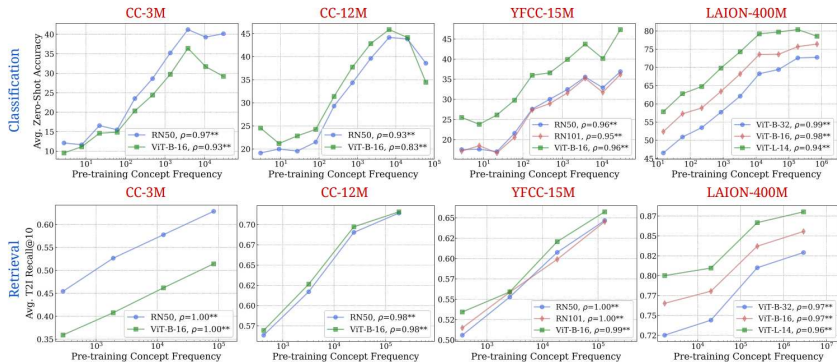


See. Oquab et al., 2023

Perspective : Foundation models ?



Perspective : Foundation models ?



See. Udandara et al, 2024

Useful ressources

- `scikit-learn` docs !

Thanks for you attention !

Let's practice !

References i

- Goodfellow, Ian, Yoshua Bengio, Aaron Courville, and Yoshua Bengio (2016). ***Deep learning***. Vol. 1. 2. MIT press Cambridge.
- Oquab, Maxime et al. (2023). “**Dinov2: Learning robust visual features without supervision**”. In: *arXiv preprint arXiv:2304.07193*.
- Ploton, Pierre et al. (2020). “**Spatial validation reveals poor predictive performance of large-scale ecological mapping models**”. In: *Nature communications* 11.1, p. 4540.
- Udandaraao, Vishaal et al. (2024). “**No zero-shot without exponential data: Pretraining concept frequency determines multimodal model performance**”. In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

References ii

- Vela, Daniel et al. (2022). **“Temporal quality degradation in AI models”**. In: *Scientific reports* 12.1, p. 11654.
- Wadoux, Alexandre MJ-C, Gerard BM Heuvelink, Sytze De Bruin, and Dick J Brus (2021). **“Spatial cross-validation is not the right way to evaluate map accuracy”**. In: *Ecological Modelling* 457, p. 109692.